



Lab Sheet 2

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice Data Manipulation Language (DML) commands for creating and modifying the structure of a table in a relational database using SQL (MySQL).

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 2: DML Commands – INSERT, UPDATE and DELETE

Perform the following SQL tasks using suitable examples:

1. Insert one record into the Student table.
2. Insert five student records using separate INSERT statements.
3. Insert multiple records into the Student table using a single statement.
4. Insert records into the Employee table.
5. Insert records into the Department table.
6. Insert records into the Product table.
7. Insert records into the Customer table.
8. Insert records into the Course table.
9. Insert a student record with values for all columns.
10. Insert a student record by specifying only selected columns.
11. Update the city of a particular student.
12. Update the salary of a particular employee.
13. Increase the salary of all employees by 10%.
14. Increase the price of all products by 5%.
15. Change the course of a particular student.
16. Update the department of an employee.
17. Update the city of all students from Meerut to Delhi.
18. Update the fees of a particular course.
19. Update multiple columns of a student simultaneously.
20. Delete a particular student record using Student_ID.
21. Delete all students belonging to a particular city.
22. Delete employees whose salary is below a specified amount.
23. Delete products whose quantity is zero.
24. Delete records satisfying multiple conditions using AND/OR.
25. Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

Instructions:

1. Implement each task in a separate query.

Submission:

1. Take screenshots of each step's output.
2. Save the SQL scripts in a text file.

Insert one record into the Student table.

```
=DROP DATABASE IF EXISTS CollegeDB;
CREATE DATABASE CollegeDB;
USE CollegeDB;
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Student_Name VARCHAR(50),
    Father_Name VARCHAR(50),
    Age INT,
    City VARCHAR(50),
    Course VARCHAR(50)
);
CREATE TABLE Employee (
    Employee_ID INT PRIMARY KEY,
    Employee_Name VARCHAR(50),
    Salary DECIMAL(10,2),
    Department VARCHAR(50)
);
CREATE TABLE Department (
    Department_ID INT PRIMARY KEY,
    Department_Name VARCHAR(50),
    Location VARCHAR(50)
);
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Price DECIMAL(10,2),
    Quantity INT
);
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(50),
    City VARCHAR(50),
    Phone VARCHAR(15)
);
CREATE TABLE Course (
    Course_ID INT PRIMARY KEY,
    Course_Name VARCHAR(50),
    Fees DECIMAL(10,2)
);

INSERT INTO Student
VALUES (101, 'Divya Verma', 'Pradeep Verma', 19, 'Saharanpur', 'B.Tech Data Science');
SELECT*FROM Student;
```

2. Insert five student records using separate INSERT statements.

```
=INSERT INTO Student VALUES
(109, 'Aman Kumar', 'Rakesh Kumar', 20, 'Delhi', 'B.Tech CSE');
INSERT INTO Student VALUES
(103, 'Priya Sharma', 'Mohan Sharma', 19, 'Meerut', 'BCA');
INSERT INTO Student VALUES
(104, 'Rahul Singh', 'Suresh Singh', 21, 'Roorkee', 'B.Tech IT');
INSERT INTO Student VALUES
(105, 'Neha Gupta', 'Raj Gupta', 20, 'Delhi', 'B.Sc CS');
INSERT INTO Student VALUES
(106, 'Karan Verma', 'Anil Verma', 19, 'Meerut', 'B.Tech CSE');
SELECT*FROM Student;
```

3. Insert multiple records into the Student table using a single statement.

```
INSERT INTO Student
(Student_ID, Student_Name, Father_Name, Age, City, Course)
VALUES
(107, 'Anjali', 'Ramesh', 19, 'Haridwar', 'B.Tech DS'),
```

```
(108, 'Vikas', 'Mahesh', 20, 'Dehradun', 'BCA'),  
(110, 'Simran', 'Rajiv', 19, 'Delhi', 'B.Tech CSE');  
SELECT * FROM Student;
```

4. Insert records into the Employee table.

```
=INSERT INTO Employee VALUES  
(201, 'Amit Sharma', 35000, 'IT');  
INSERT INTO Employee VALUES  
(202, 'Riya Gupta', 42000, 'HR');  
INSERT INTO Employee VALUES  
(203, 'Rohit Singh', 28000, 'Sales');  
INSERT INTO Employee VALUES  
(204, 'Neha Verma', 50000, 'Finance');  
select*from employee;
```

5. Insert records into the Department table.

```
=INSERT INTO Department VALUES  
(1, 'IT', 'Delhi');  
INSERT INTO Department VALUES  
(2, 'HR', 'Roorkee');  
INSERT INTO Department VALUES  
(3, 'Finance', 'Haridwar');  
INSERT INTO Department VALUES  
(4, 'Sales', 'Meerut');  
select*from department;
```

6. Insert records into the Product table.

```
=INSERT INTO Product VALUES  
(301, 'Laptop', 55000, 10);  
INSERT INTO Product VALUES  
(302, 'Mouse', 500, 25);  
INSERT INTO Product VALUES  
(303, 'Keyboard', 1000, 15);  
INSERT INTO Product VALUES  
(304, 'Monitor', 12000, 5);  
INSERT INTO Product VALUES  
(305, 'Printer', 8000, 0);  
select*from product;
```

7. Insert records into the Customer table.

```
=INSERT INTO Customer VALUES  
(401, 'Rahul Kumar', 'Delhi', '9876543210');  
INSERT INTO Customer VALUES  
(402, 'Priya Singh', 'Meerut', '9876543211');  
INSERT INTO Customer VALUES  
(403, 'Ankit Sharma', 'Roorkee', '9876543212');  
INSERT INTO Customer VALUES  
(404, 'Neha Gupta', 'Haridwar', '9876543213');  
select*from customer;
```

8. Insert records into the Course table.

```
=INSERT INTO Course VALUES  
(501, 'B.Tech Data Science', 90000);  
INSERT INTO Course VALUES  
(502, 'B.Tech CSE', 85000);  
INSERT INTO Course VALUES  
(503, 'BCA', 60000);  
INSERT INTO Course VALUES  
(504, 'B.Sc Computer Science', 55000);  
select * from course;
```

9. Insert a student record with values for all columns.

```
=INSERT INTO Student
(Student_ID, Student_Name, Father_Name, Age, City, Course)
VALUES
(110, 'Arjun Mehta', 'Vijay Mehta', 20, 'Chandigarh', 'B.Tech CSE');
```

10. Insert a student record by specifying only selected columns.

```
=INSERT INTO Student
(Student_ID, Student_Name, City)
VALUES
(111, 'Pooja Sharma', 'Delhi');
```

11. Update the city of a particular student.

```
=UPDATE Student
SET City = 'Delhi'
WHERE Student_ID = 101;
SET SQL_SAFE_UPDATES = 1;
SELECT*FROM STUDENT;
```

12. Update the salary of a particular employee.

```
=UPDATE Employee
SET Salary = 40000
WHERE Employee_ID = 201;
SET SQL_SAFE_UPDATES=1;
SELECT*FROM EMPLOYEE;
```

13. Increase the salary of all employees by 10%.

```
=UPDATE Employee
SET Salary = Salary * 1.10;
select*from employee;
```

14. Increase the price of all products by 5%.

```
=UPDATE Product
SET Price = Price * 1.05;
select * from product;
```

15. Change the course of a particular student.

```
=UPDATE Student
SET Course = 'B.Tech Data Science'
WHERE Student_ID = 103;
select*from course;
```

16. Update the department of an employee.

```
=UPDATE Employee
SET Department = 'IT'
WHERE Employee_ID = 203;
select*from employee;
```

17. Update the city of all students from Meerut to Delhi.

```
=UPDATE Student
SET City = 'Delhi'
WHERE City = 'Meerut';
select*from student;
```

18. Update the fees of a particular course.

```
=UPDATE Course
SET Fees = 65000
```

```
WHERE Course_ID = 503;
select*from student;
```

19. Update multiple columns of a student simultaneously.

```
=UPDATE Student
SET Age = 21,
    City = 'Haridwar',
    Course = 'B.Tech Data Science'
WHERE Student_ID = 102;
select*from student;
```

20. Delete a particular student record using Student_ID.

```
=DELETE FROM Student
WHERE Student_ID = 111;
select*from student;
```

21. Delete all students belonging to a particular city.

```
=DELETE FROM Student
WHERE City = 'Delhi';
select*from student;
```

22. Delete employees whose salary is below a specified amount.

```
=DELETE FROM Employee
WHERE Salary < 30000;
select*from employee;
```

23. Delete products whose quantity is zero.

```
=DELETE FROM Product
WHERE Quantity = 0;
select*from product;
```

24. Delete records satisfying multiple conditions using AND/OR.

```
=DELETE FROM Employee
WHERE Salary < 35000
AND Department = 'Sales';
select*from employee;
```

25. Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

```
=INSERT INTO Student
VALUES(120, 'Kavya Sharma', 'Raj Sharma', 19, 'Roorkee', 'B.Tech CSE');
UPDATE Student
SET City = 'Delhi'
WHERE Student_ID = 120;
SELECT * FROM Student;
UPDATE Student
SET Course = 'B.Tech Data Science'
WHERE Student_ID = 120;
SELECT * FROM Student;
UPDATE Student
SET Age = 20,
    City = 'Haridwar'
WHERE Student_ID = 120;
SELECT * FROM Student;
DELETE FROM Student
WHERE Student_ID = 120;
SELECT * FROM Student;
```

	Student_ID	Student_Name	Father_Name	Age	City	Course
▶	101	Divya Verma	Pradeep Verma	19	Saharanpur	B.Tech Data Science
•	NULL	NULL	NULL	NULL	NULL	NULL

	Employee_ID	Employee_Name	Salary	Department
▶	201	Amit Sharma	35000.00	IT
	202	Riya Gupta	42000.00	HR
	203	Rohit Singh	28000.00	Sales
	204	Neha Verma	50000.00	Finance
•	NULL	NULL	NULL	NULL

	Course_ID	Course_Name	Fees
	501	B.Tech Data Science	90000.00
	502	B.Tech CSE	85000.00
	503	BCA	60000.00
	504	B.Sc Computer Science	55000.00
•	NULL	NULL	NULL

	Customer_ID	Customer_Name	City	Phone
▶	401	Rahul Kumar	Delhi	9876543210
	402	Priya Singh	Meerut	9876543211
	403	Ankit Sharma	Roorkee	9876543212
	404	Neha Gupta	Haridwar	9876543213
•	NULL	NULL	NULL	NULL

Result Grid						
Filter Rows:						
	Student_ID	Student_Name	Father_Name	Age	City	Course
▶	101	Divya Verma	Pradeep Verma	19	Saharanpur	B.Tech Data Science
	103	Priya Sharma	Mohan Sharma	19	Meerut	BCA
	104	Rahul Singh	Suresh Singh	21	Roorkee	B.Tech IT
	105	Neha Gupta	Raj Gupta	20	Delhi	B.Sc CS
	106	Karan Verma	Anil Verma	19	Meerut	B.Tech CSE
	107	Anjali	Ramesh	19	Haridwar	B.Tech DS
	108	Vikas	Mahesh	20	Dehradun	BCA
	109	Aman Kumar	Rakesh Kumar	20	Delhi	B.Tech CSE
	110	Simran	Rajiv	19	Delhi	B.Tech CSE
	111	Pooja Sharma	NULL	NULL	Delhi	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL

	Product_ID	Product_Name	Price	Quantity
▶	301	Laptop	55000.00	10
	302	Mouse	500.00	25
	303	Keyboard	1000.00	15
	304	Monitor	12000.00	5
	305	Printer	8000.00	0
•	NULL	NULL	NULL	NULL

	Student_ID	Student_Name	Father_Name	Age	City	Course
▶	111	Pooja Sharma	NULL	NULL	Delhi	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL